

Problem Set 2

ECON 2010 – Macroeconomics

Spring 2018

Due at the start of class Week 5, February 14. You have two weeks to complete the problem set. No late problem sets will be accepted. Please refer to the syllabus section on late assignments.

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct the problem set together in class on the 14th.

Name:

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 (55 points)

In this problem, you'll consider the supply of and demand for kale. This probably isn't something you consider often. I certainly don't. But let's see how it goes. If you don't already know, kale is a leafy green that has become quite popular in the last five years. Between 2012 and 2013, sales of kale grew 40% in the US; remarkable for a vegetable. Kale has been called a superfood — because it's high in vitamins and minerals — and a fad food — because at any new or trendy restaurant you go to, you'll inevitably find a kale salad on the menu. Some people think kale is pretty tasty. Some people think everyone who considers kale tasty has lost their mind. But we won't get into that debate.

Instead, consider a market for kale that includes just two consumers — Darius and Georgio — and two producers (growers) — MegaKale Corp. and Kale Yeah!. The demand schedules for each

Darius and Georgio are given below, as well as the supply schedules for each of the two producers.

Table 1: Individual demand schedules for kale

Price (per oz.)	Darius' Quantity Demanded (oz.)	Georgio's Quantity Demanded (oz.)
\$4.00	0	0
\$3.00	5	2
\$2.00	10	4
\$1.00	15	6
\$0.00	20	8

Table 2: Individual supply schedules for kale

Price (per oz.)	MegaKale Corp.'s Quantity Supplied (oz.)	Kale Yeah!'s Quantity Supplied (oz.)
\$4.00	12	8
\$3.00	9	6
\$2.00	6	4
\$1.00	3	2
\$0.00	0	0

- (a) Restricting the example to just two consumers and two producers is an example of employing which fundamental principle of economic thinking — that is, which tool from the economists toolbox? **(1 point) Remember: Don't write on this problem set. See the problem set instructions.**

- (b) Draw Darius's individual demand curve on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the demand curve is downward sloping. Be

sure to label each of your axes. **(3 points)**

- (c) Draw Georgio's individual demand curve on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the demand curve is downward sloping. Be sure to label each of your axes. **(3 points)**

- (d) We've seen in class that we can apply the principle of marginal thinking to tradeoffs between two variables. For example, we saw with the production possibilities frontier (PPF) that the slope of a given line segment on the frontier corresponds to the *marginal* opportunity cost (also sometimes referred to as the *marginal rate of transformation*) of producing one good in terms of the other good.

We can also apply the principle of marginal thinking to the price/quantity tradeoff that a given consumer or producer faces. These marginal quantities are studied at great length in microeconomics, and because of that they have their own special names. The marginal tradeoff between price and quantity for consumers is called the *price elasticity of demand*, while the marginal tradeoff between price and quantity for producers is called the *price elasticity of supply*. These elasticities tell us something about how *sensitive* different consumers, producers, and markets are to changes in prices. Using the same strategy that we used to find the marginal opportunity cost for a line segment on the PPF, calculate the *price elasticity of*

demand for **each** Darius and Georgio. **(10 points total)**

(1) What is Darius' price elasticity of demand? **(4 points)**

(2) What is Georgio's price elasticity of demand? **(4 points)**

(3) Given these two elasticities, who is *more* sensitive to price — Darius, or Georgio?
That is, for a given *increase* in price, who will demand *less* kale? **(2 points)**

(e) If Darius and Georgio are the only two consumers in the market for kale, draw the market demand curve for kale on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the demand curve is downward sloping. Be sure to label each of your axes. **(5 points)**

(f) Draw the individual supply curve for MegaKale Corp. on an x,y plane. Choose your axes as

chosen in class (and as demonstrated in your textbook) such that the supply curve is upward sloping. Be sure to label each of your axes. **(3 points)**

- (g) Draw the individual supply curve for Kale Yeah! on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the supply curve is upward sloping. Be sure to label each of your axes. **(3 points)**

- (h) If MegaKale Corp. and Kale Yeah! are the only two producers in the market for kale, draw the market supply curve for kale on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the supply curve is upward sloping. Be sure to label each of your axes. **(5 points)**

- (i) If MegaKale Corp. produces conventional (non-organic) kale and Kale Yeah! produces organic kale would you keep the market supply curve the same as in part (h) above? Or would you draw a different market supply somehow. Do **not** draw or re-draw the market supply. Just state whether it would be the same or different. Then give 1 or *at most* 2 sentences to explain why it would stay the same or be different. **(2 points)**
- (j) Now that you've constructed the market demand curve (part (e)) and the market supply curve (part (h)) for kale, draw each together on new x,y plane. Again, remember to label your axes. Use a dotted line from the intersection point of the two curves to indicate the *approximate* location of the equilibrium quantity. Do the same for equilibrium price. (Note: In order to calculate the equilibrium price and quantity, you would need to use algebra to solve a system of equations. I'm not asking you to do that.) Is the equilibrium price more or less than \$1.00? Is the equilibrium quantity more or less than 16? (Note: Draw your supply and demand curves carefully to answer these questions.) **(5 points)**
- (k) Consider the market equilibrium for kale that you found in part (j) above. Would, *all else equal* (a.k.a. ceteris parabis, holding everything else fixed), Kale Yeah! be better off (that is, would the producer would make more money) if it raised its price by \$0.10 per pound? Explain in just a few sentences why or why not. Note: You do not need to do any math to answer this question. **(5 points)**

- (l) Consider the market equilibrium for kale that you found in part (j) above. Re-draw that equilibrium. Then, draw a new market equilibrium on the same plot to reflect a news report that says kale is *not* as much of a superfood as previously thought. It doesn't have the quantities of vitamins and minerals that nutritionists used to think it had. Assume that a large part of the reason people were consuming kale was for its value as a super food.

One of your two curves (either market demand or market supply) will stay fixed in your new plot. The other curve will shift. Draw the new shifted curve with a dotted line. (Use solid lines for the starting equilibrium; i.e. the equilibrium from part (j)). I don't tell you exactly how much the curve should shift, so just draw a reasonable shift given the information above. The important part is the direction of the shift relative to the curve's starting point. Is the new equilibrium price for kale higher or lower than it was in part (j)? Is the new equilibrium quantity higher or lower than it was in part (j)? **(5 points)**

- (m) Consider the market equilibrium for kale that you found in part (j) above. Re-draw that equilibrium. Then, draw a new market equilibrium on the same plot to reflect an increase in the price of the fertilizer that both MegaKale Corp. and Kale Yeah! use to fertilize their crops.

One of your two curves (either market demand or market supply) will stay fixed in your new plot. The other curve will shift. Draw the new shifted curve with a dotted line. (Use solid

lines for the starting equilibrium; i.e. the equilibrium from part (j)). I don't tell you exactly how much the curve should shift, so just draw a reasonable shift given the information above. The important part is the direction of the shift relative to the curve's starting point. Is the new equilibrium price for kale higher or lower than it was in part (j)? Is the new equilibrium quantity higher or lower than it was in part (j)? **(5 points)**

- (n) Consider the following statement: "Because kale is such a healthy vegetable, the government should adopt a price ceiling for kale so that it never becomes too expensive and people will have access to a cheap and healthy vegetable in the grocery store." Is this an example of a positive or a normative statement? **(1 point)**

Problem 2 (16 points)

In this problem you'll consider the components of an economy similar to South Africa in 2016. (I could not find all of the component values of GDP for South Africa, so some of these numbers are approximations.)

Table 3: Components of an example economy similar to South Africa, in billions of \$

Consumption	Gross Private Investment	Private Depreciation	Government Purchases	Government Transfers	Exports	Imports
226	340	291	89	60	70	74

- (a) Use the above table to calculate private investment expenditures (i.e. net investment) for the economy. **(2 points)**

(b) Use the above table to calculate total government purchases for the economy. **(2 points)**

(c) Use the above table to calculate net exports for the economy. **(2 points)**

(d) Gross Domestic Product (GDP) for a country is calculated as the following: $GDP = consumption + investment + government purchases + net exports$. Use the table above and the calculations that you've made in parts (a)-(c) to calculate GDP for this economy. **(2 points)**

(e) If the population of this country similar to South Africa is 50 million, what is average GDP per person — also known as *GDP per capita*? **(3 points)**

(f) Does the GDP per capita number that you found in part (e) above mean that every person in South Africa is earning that exact income? Yes or no. **(1 point)**

(g) Let's now think prices and the goods produced in our example economy. Let's say that there are three things (goods and services) produced in the economy — diamonds, gold, and safari trips. The quantities of these goods produced, and their prices are listed in the table below. Diamonds are measured in millions of carats (a unit of weight), gold is measured in millions metric tons, and safaris are measured in millions of trips.

Table 4: Production and prices in an economy similar to South Africa

Year	Quantity Produced (millions)			Price		
	Diamonds	Gold	Safaris	Diamonds	Gold	Safaris
2015	10	150	1	\$1,300	\$2,000	\$10,000
2016	15	120	0.7	\$1,350	\$2,000	\$12,000

Calculate nominal GDP for each of the two years listed above. **(3 points)**

- (h) Using the table in part (g) above, calculate real GDP in 2016. Did real GDP increase between 2015 and 2016? **(4 points)**

Problem 3 (21 points)

For each of the statements below, do **both** following. First, indicate *which* component or components of GDP are related to the statement (your answer will be either consumption, private investment, government purchases, or net exports) (worth 1 point). Second, indicate how US GDP is affected (your answer will either be increases, decreases, stays the same, or unknown) (worth 2 points).

For each of the statements, use the principle of *all else equal* (a.k.a. ceteris parabis, holding other factors fixed) to conduct your analysis and answer the questions.

- (a) Households in the US go on a binge of buying and eating US-produced popcorn. **(3 points)**
- (b) Amelie, who is a Canadian citizen who lives in Canada, buys a new car in Michigan, that was built by Ford Motors in Michigan. **(3 points)**
- (c) US Steel Corporation purchases a new \$10 million steel rolling machine for its factory. The steel rolling machine was produced in Japan. **(3 points)**

(d) Nashville State Community College, a public school, builds a new library for \$5 million. **(3 points)**

(e) Barry sells his 15 year old Toyota Tercel to a neighbor. **(3 points)**

(f) Your best friend spends 3 hours fixing the broken wheel on your car for free (wow, what a nice friend); a job that would have cost you \$300 at your local service station. **(3 points)**

Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.